

Practical Laboratory 2

OPNsense Firewall Policy Testing Logging and Packet Analysis

WADF105 Network Security Fundamentals

In this laboratory we will use the commissioned network containing icdfa-nslab-client-v1 and icdfa-nslab-firewall-v1 to create controlled firewall rules and verify their effect from the Ubuntu client, examine OPNsense log evidence, and correlate the results with a packet capture.

Learning objectives

- Explain how top-down firewall rule processing affects traffic.
- Create and test a narrow rule without accidentally blocking all connectivity.
- Use protocol, source, destination and port fields correctly.
- Read OPNsense firewall logs and identify the rule that blocked a packet.
- Correlate firewall decisions with Ubuntu packet-capture evidence.
- Disable test rules and verify that normal connectivity is restored.

Part A Record the baseline

1. Started icdfa-nslab-firewall-v1 first, then start icdfa-nslab-client-v1.
2. On icdfa-nslab-client-v1, record the current IPv4 address and default route.

```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo                UNKNOWN          127.0.0.1/8
enp0s3            UP                10.10.10.102/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto dhcp src 10.10.10.102 metric 100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.102 metric 100
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=45.6 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=49.1 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=65.9 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=51.6 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3008ms
rtt min/avg/max/mdev = 45.636/53.071/65.938/7.726 ms
icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::ac42:93f3 example.com
2606:4700:10::6814:179a example.com
icdfa@icdfa-nslab-client-v1:~$ curl -I http://example.com
HTTP/1.1 200 OK
Date: Thu, 01 Oct 2026 08:56:21 GMT
Content-Type: text/html; charset=utf-8
Connection: keep-alive
Server: cloudflare
Last-Modified: Mon, 28 Sep 2026 16:19:32 GMT
Allow: GET, HEAD
Accept-Ranges: bytes
Age: 1516
cf-cache-status: HIT
CF-RAY: a43a40ec6a9837b1-KHI
icdfa@icdfa-nslab-client-v1:~$
```

Part B Review rule order

3. In OPNsense, open Firewall, Rules, LAN.
4. Locate the Default allow LAN to any rule. This broad allow rule permits icdfa-nslab-client-v1 to reach external destinations.

Part C Blocked ICMP to one test address

Field	Required value
Action	Block
Quick	Enabled if displayed
Interface	LAN
Direction	In
TCP IP Version	IPv4
Protocol	ICMP
ICMP type	Any or Echo request
Source	LAN net
Destination	Single host or Network: 1.1.1.1/32
Log packets handled by this rule	Enabled
Description	LAB2 BLOCK ICMP TO 1.1.1.1

The screenshot displays the OPNsense Firewall Rules configuration interface. The 'LAN' interface is selected. The rule list shows the following rules:

Enabled	Icon	Version	Protocol	Source	Port	Destination	Port	Gateway	Description	Commands
☐		LAN	IPv4	ICMP	LAN network	*	1.1.1.1/32	*	LAB2 BLOCK ICMP TO 1.1.1.1	
☐		LAN	IPv4	TCP	LAN network	*	*	http	Default allow LAN to any rule	
☐		LAN	IPv4	*	LAN network	*	*	*	Default allow LAN to any rule	
☐		LAN	IPv6	*	LAN network	*	*	*	Default allow LAN IPv6 to any rule	

The bottom of the screen shows an 'Apply' button and a message: 'After changing settings, please remember to apply them.'

The ping timed out or report 100 percent packet loss. Now confirm that DNS and HTTPS still work.

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3167ms

icdfa@icdfa-nslab-client-v1:~$ getent hosts example.com
2606:4700:10::ac42:93f3 example.com
2606:4700:10::6814:179a example.com
icdfa@icdfa-nslab-client-v1:~$ curl -I https://example.com
HTTP/2 200
date: Thu, 01 Oct 2026 17:13:14 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Mon, 28 Sep 2026 16:19:32 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 3
cf-cache-status: HIT
cf-ray: a43d18cb1f58ce6a-SIN

icdfa@icdfa-nslab-client-v1:~$
```

Part D Blocked outbound HTTP while allowing HTTPS

Create a second temporary rule above the broad allow rule. This rule will block unencrypted web traffic on TCP port 80 but leave HTTPS on TCP port 443 available.

Field	Required value
Action	Block
Interface	LAN
Direction	In
TCP IP Version	IPv4
Protocol	TCP
Source	LAN net
Destination	Any
Destination port range	HTTP to HTTP, port 80

Field	Required value
Log packets handled by this rule	Enabled
Description	LAB2 BLOCK OUTBOUND HTTP

On icdfa-nslab-client-v1, compare an HTTP request with an HTTPS request.

```
curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
```

```
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I http://example.com
curl: (28) Connection timed out after 10002 milliseconds
icdfa@icdfa-nslab-client-v1:~$ curl --max-time 10 -I https://example.com
HTTP/2 200
date: Thu, 01 Oct 2026 17:15:13 GMT
content-type: text/html; charset=utf-8
server: cloudflare
last-modified: Mon, 28 Sep 2026 16:19:32 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 5
cf-cache-status: HIT
cf-ray: a43d1bb3d80dfdc7-SIN
icdfa@icdfa-nslab-client-v1:~$
```

Part E Examine firewall logs

- Opened Firewall, Log Files, Live View in OPNsense.
- Generated new blocked traffic by repeating the ping and HTTP curl commands from icdfa-nslab-client-v1.
- Filter the log by the icdfa-nslab-client-v1 IP address, destination 1.1.1.1, destination port 80, or the LAB2 rule description.
- Opened a matching entry and record the timestamp, interface, source IP, destination IP, protocol, destination port where applicable, action and rule label.

18:20:43	ICMP	10.10.10.102	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
18:20:42	ICMP	10.10.10.102	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
18:20:40	ICMP	10.10.10.102	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
18:20:39	ICMP	10.10.10.102	1.1.1.1	block	LAB2 BLOCK ICMP TO 1.1.1.1
18:20:38	TCP	10.10.10.102:55000	1.1.1.1:80	block	LAB2 BLOCK OUTBOUND HTTP

Part F Correlate with Wireshark

- Open Wireshark on Ubuntu and capture on the active 10.10.10.x interface.
- Repeat ping -c 4 1.1.1.1 and curl --max-time 10 -I http://example.com.
- Stop the capture.
- Use the filters below to isolate the traffic.

Display filter	Expected observation
icmp && ip.addr == 1.1.1.1	Echo requests leave Ubuntu, but no echo replies return.
tcp.dstport == 80	TCP SYN packets may be retransmitted because the firewall silently blocks them.
tcp.port == 443	A permitted HTTPS connection shows a completed TCP handshake and encrypted TLS traffic.

The image shows a Wireshark packet capture interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons for packet capture and analysis. The main display area shows a list of captured packets with columns for No., Time, Source, Destination, Protocol, Length, and Info. The filter bar at the top of the packet list is set to 'tcp.dstport == 80 || tcp.port == 443 || icmp'. The packet list shows several packets, including ICMP echo requests and TCP SYN packets. The detailed view pane at the bottom shows the selected packet (No. 14) and its structure, including Ethernet II, Internet Protocol Version 4, and Hypertext Transfer Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.10.10.1	10.10.10.1	HTTP	672	GET /api/diagnostics/
4	0.187707170	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
6	0.189846834	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
7	1.361599924	10.10.10.102	91.189.91.58	TCP	74	35528 → 80 [SYN] Seq=
8	1.482055829	10.10.10.102	10.10.10.1	HTTP	672	GET /api/diagnostics/
11	1.731979335	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
13	1.734968803	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
14	2.406231582	10.10.10.102	91.189.91.58	TCP	74	[TCP Retransmission]
15	2.481396578	10.10.10.102	10.10.10.1	HTTP	672	GET /api/diagnostics/
18	2.785668444	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
20	2.787425127	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
21	3.433002687	10.10.10.102	91.189.91.58	TCP	74	[TCP Retransmission]
22	4.454453214	10.10.10.102	91.189.91.58	TCP	74	[TCP Retransmission]
23	5.213602602	10.10.10.102	10.10.10.1	HTTP	672	GET /api/diagnostics/
25	5.479043529	10.10.10.102	91.189.91.58	TCP	74	[TCP Retransmission]
27	5.517765365	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
29	5.518344216	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
31	5.522545103	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=
33	5.528561825	10.10.10.102	10.10.10.1	TCP	66	59802 → 80 [ACK] Seq=

Frame 1: 672 bytes on wire (5376 bits), 672 bytes captured (5376 bits) on interface eth0
 Ethernet II, Src: PCSSystemtec_35:d0:7a (08:00:27:00:00:00), Dst: 08:00:27:00:00:00
 Internet Protocol Version 4, Src: 10.10.10.102, Destination: 91.189.91.58
 Transmission Control Protocol, Src Port: 59802, Destination Port: 80
 Hypertext Transfer Protocol

Part G Inspect states and automatic NAT

- Open Firewall, Diagnostics, States and filter by the icdfa-nslab-client-v1 IP address.
- Generated a fresh HTTPS request from icdfa-nslab-client-v1 and refresh the state table.

15. Opened Firewall, NAT, Outbound and confirm the current mode.

Firewall: Diagnostics: States

States Actions									
Q Search ↺ 50									
☐	Int	Dir	Proto	Source	Nat	Destination	Gateway	Drop	Rule
☐	all	←	icmp	10.0.2.15:18198		10.0.2.2:8		0:0	allow access to ...
☐	all	←	icmp	10.0.2.15:18926		172.16.100.1:8		0:0	anti-lockout rule
☐	all	→	tcp	10.10.10.102:33...		34.107.243.93:443		ESTABLISHED:E...	LAB2 BLOCK IC...
☐	all	←	tcp	10.0.2.15:11177	10.10.10.102:33...	34.107.243.93:443		ESTABLISHED:E...	let out anything...
☐	all	→	tcp	10.10.10.102:54...		10.10.10.1:80		FIN_WAIT_2:FIN...	anti-lockout rule
☐	all	→	tcp	10.10.10.102:46...		10.10.10.1:80		FIN_WAIT_2:FIN...	anti-lockout rule
☐	all	→	tcp	10.10.10.102:43...		151.101.141.91:...		ESTABLISHED:E...	Default allow L...
☐	all	←	tcp	10.0.2.15:49820	10.10.10.102:43...	151.101.141.91:...		ESTABLISHED:E...	let out anything...
☐	all	→	tcp	10.10.10.102:43...		151.101.141.91:...		FIN_WAIT_2:FIN...	Default allow L...
☐	all	←	tcp	10.0.2.15:42658	10.10.10.102:43...	151.101.141.91:...		FIN_WAIT_2:FIN...	let out anything...
☐	all	→	tcp	10.10.10.102:53...		10.10.10.1:80		FIN_WAIT_2:FIN...	anti-lockout rule

Part H Restore the laboratory

16. Return to Firewall, Rules, LAN.

17. Disable the two rules named LAB2 BLOCK ICMP TO 1.1.1.1 and LAB2 BLOCK OUTBOUND HTTP. Do not delete them until your screenshots are complete.

18. Apply the changes.

19. If old connections affect the test, clear only the relevant states or wait briefly, then test again.

```
ping -c 4 1.1.1.1
curl --max-time 10 -I http://example.com
curl --max-time 10 -I https://example.com
```

Firewall: Rules

The screenshot shows the Firewall Rules configuration interface. At the top, there are filters for 'All rules', 'Categories', and a search bar. Below these are controls for refreshing, a limit of 500, a list view, and a refresh icon. The main table lists rules with columns for checkboxes, icons, names, interfaces, versions, protocols, sources, ports, destinations, and actions. The 'Interface rules' section is expanded, showing three rules. The first rule is disabled, the second is disabled, and the third is enabled. The third rule is a block rule for LAN traffic on port 80.

☐		☒	Icons		Version	Protocol	Source	Port	Des	Commands	
☐		☐			LAN	IPv4	ICMP	LAN network	*	1.1.1.1	
☐		☐			LAN	IPv4	TCP	LAN network	*	*	
☐		☒			LAN	IPv4	*	LAN network	*	*	

Troubleshooting guide

Problem	Likely cause	Corrective action
Traffic is still allowed	The block rule is below the allow rule, uses the wrong interface, or changes were not applied	Move the rule above Default allow LAN to any, confirm LAN, and select Apply changes.
All traffic stops	The rule is broader than intended	Disable it immediately. Recheck protocol, destination and port before re-enabling.
No log entries appear	Logging was not enabled or the filter is too narrow	Edit the rule and enable logging; clear the Live View filter and generate fresh traffic.
HTTP test succeeds	Browser or command followed a redirect, or the rule targets the wrong port	Use curl without -L, verify TCP destination port 80, and generate a new connection.
Access does not return after disabling rules	Existing firewall states or another configuration problem	Apply changes, clear only relevant states, repeat Laboratory 1 baseline checks, and verify the rules are disabled.

CONCLUSION:

This practical laboratory successfully demonstrated the implementation, evaluation, and packet-level analysis of controlled firewall policies using the OPNsense gateway (icdfa-nslab-firewall-v1) and an Ubuntu client workstation (icdfa-nslab-client-v1). The exercise verified that OPNsense evaluates interface rules sequentially from top to bottom, confirming that specific block rules must be placed above broad allow rules to effectively intercept targeted traffic. By configuring narrow rules to block ICMP traffic to 1.1.1.1 and outbound HTTP traffic on TCP port 80, the laboratory proved that granular filtering can restrict unencrypted web traffic and diagnostic pings without disrupting core network services like DNS name resolution and encrypted HTTPS on port 443. Furthermore, correlation between OPNsense Live View logs and raw packet captures in Wireshark highlighted the underlying mechanics of stateful filtering. Silently dropped TCP SYN packets for port 80 resulted in client-side connection timeouts and retransmissions, while permitted HTTPS connections successfully formed active entries in the firewall state table, completing full TCP and TLS handshakes. Additionally, inspecting the firewall states confirmed that Automatic Outbound NAT effectively translated the client's internal private IP address (10.10.10.0/24) to the public WAN gateway address for external access. Finally, disabling the test rules and confirming restored full connectivity satisfied essential security laboratory hygiene practices, returning the network to a known-good operational baseline.